ANIMAL NUTRITION

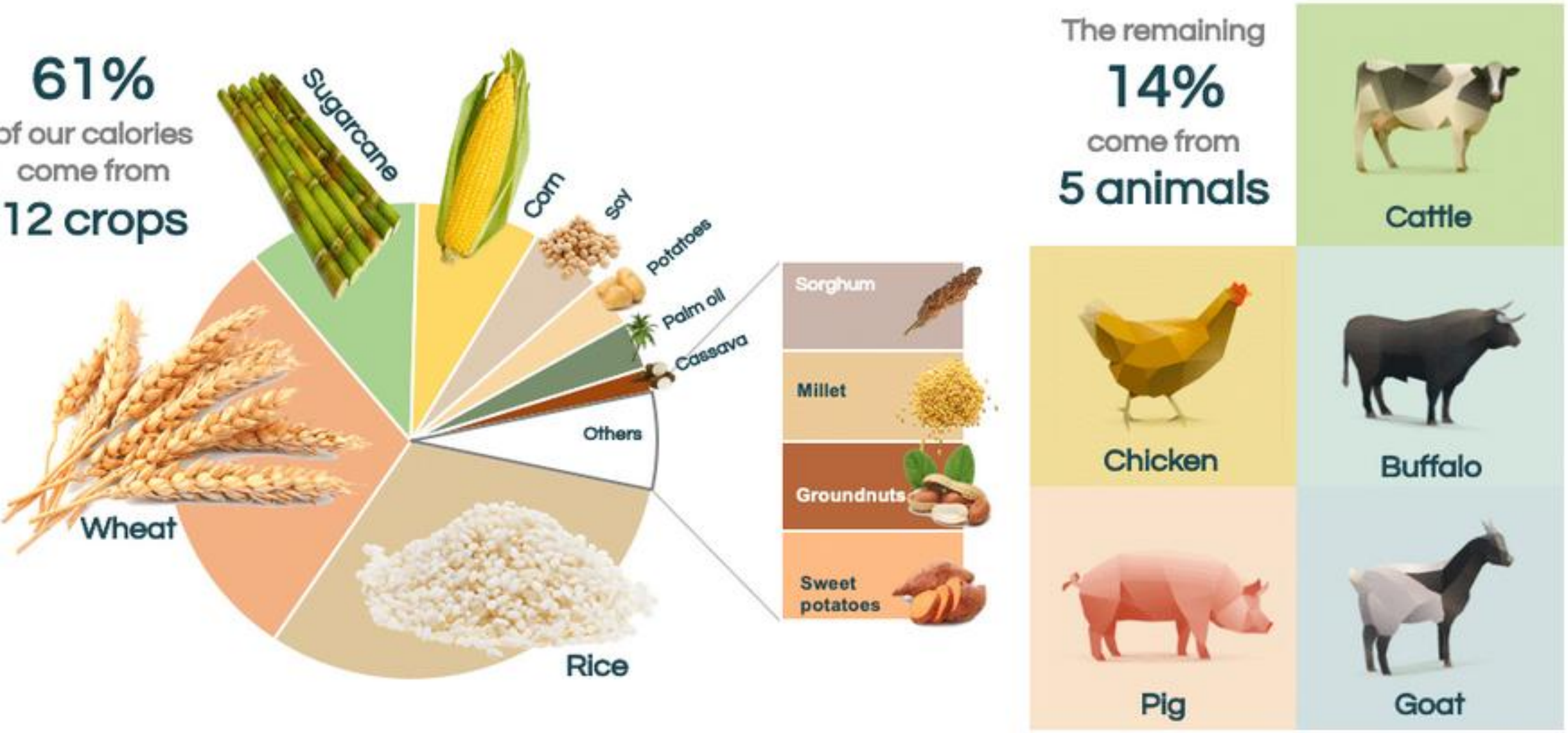
- Introduction
- Concepts of nutrition
- Role in modern agriculture & society
- Trends in consumption of animal products
- Improved feed efficiency
- Dry matter & Dry matter intake

Lecture 1: Introduction to Animal Nutrition



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World Food System Support



INTRODUCTION

“A Hungry People Listens not to Reason nor are its Demands turned aside by Prayer”

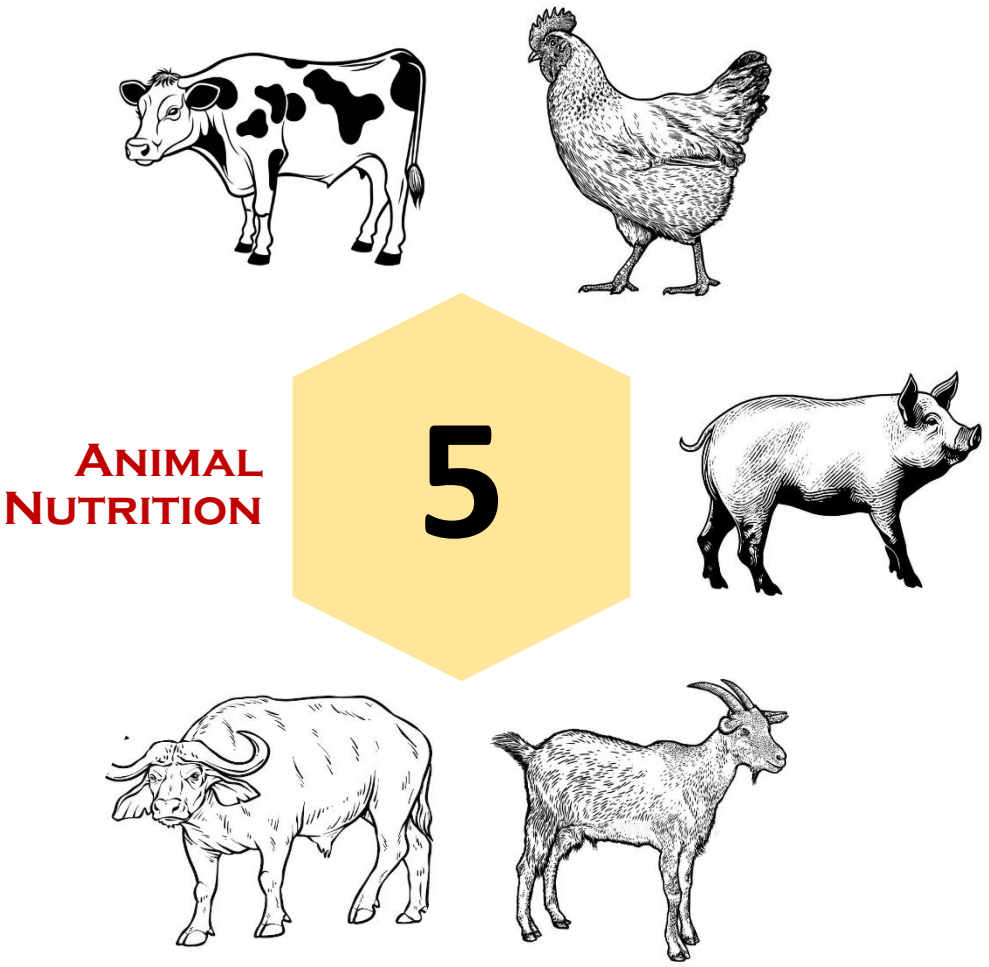
Hunger impacts human health & sustainable development.

The study of how dietary nutrients are ingested, digested, absorbed, transported, metabolized and excreted by animals is animal nutrition.

Encompasses quantitative (how much), qualitative (what types), and functional (how nutrients work) aspects.

A nutrient is any chemical element/ compound in a diet that maintains life and support production.

Animal Nutrients are always contained in the dry matter of feeds.



INTRODUCTION

Essential/ Indispensable Nutrients: required in diets because animals cannot synthesize them in sufficient amounts to meet metabolic needs.

Plants provide an essential nutritional link between soil and animal life. They photosynthesize and form nutrients that are available to animals.

Animals transform the plant products of little value into nutritious human food.

Therefore, proper nutrition leads to:

- Production of milk, eggs, meat, wool, etc.
- Production of manure for soil amendments and sustainable plant growth.

HISTORY

Prehistoric & Ancient Times

- Early herders learned by observating which plants made animals thrive.
- Traditional Knowledge: “let animals graze after rain,” “supplement with salt licks” passed down orally.
- Early Writings: Greek and Roman texts discussed fodder types and seasonal grazing.
- Chinese classics (like the “Book of Agriculture” by Jia Sixie, 6th c) outlined forage crops and basic feeding regimes.

Renaissance to 18th Century

- Systematic Observation: European agricultural began noting growth differences on different forages.
- Quantitative Records: Some estate managers kept rudimentary feed and weight records, but underlying mechanisms remained mysterious.

HISTORY

19th Century: Birth of Scientific Nutrition

- ❑ Proximate Analysis (Weende System, 1860s): Partition feed into MC, ash, CP, fat, fiber, etc.
- ❑ Analytical framework linking chemical composition to animal growth.
- ❑ Energy Concepts: Introduction of GE & DE

Early–Mid 20th Century: Refinement & Expansion

- ❑ Vitamin Discovery: essential vitamins (A, D, B complex)
- ❑ Mineral Nutrition: Recognition of trace minerals (e.g., copper, selenium) and macro-minerals (Ca, P).
- ❑ Ration Formulation: Development of NRC nutrient requirement tables

1960s–1980s: Fiber & Energy Systems

- ❑ Van Soest Fiber Analysis: Introduction of NDF & ADF to better predict forage digestibility in ruminants.
- ❑ Net Energy Systems: Partitioning energy into maintenance, growth, lactation components for precise diet formulation.

1990s–2000s: Molecular & "Omics" Era

- ❑ Amino Acid Balancing: Shift from CP to ideal AA profiles (e.g., lysine, methionine) for monogastrics.
- ❑ Genomics & Proteomics: Early studies on genetic markers for feed efficiency and gut-microbiome interactions.

2010s–Present: Precision & Sustainability

- ❑ Precision Feeding Technologies
- ❑ Automated feeders, electronic ID tags, real-time intake monitoring.
- ❑ Models that adjust rations daily based on performance, weather, even satellite-based forage maps.
- ❑ Alternative Feed Ingredients: Insect meals, algae, single-cell proteins as novel, low-footprint nutrient sources.
- ❑ Environmental Focus: Life-cycle assessments to minimize greenhouse gases, nutrient runoff; formulation for methane reduction (e.g., seaweed supplements).

Ancient Period

Animals thrive when allowed to graze this particular pasture

Renaissance/ Early Modern Period

Animals grazing on this forage appear to grow better; let's observe, compare and record the difference

18th – 19th Century

Let's measure the feed, animal weight and production and investigate what is actually contained in the feed

Modern Nutrition

Let's analyze nutrients chemically and formulate diets according to the animal's precise requirements

1. Enhancing Animal Health and Welfare

- ❑ Strengthens immunity,
- ❑ Reducing disease incident
- ❑ Prevent metabolic disorders improves

2. Improving Livestock Productivity

- ❑ Meat, milk, egg, and wool yields increase when physiological needs are matched with nutrition.
- ❑ Precision nutrition allows farmers to optimize FCR
- ❑ Enhances product quality: better marbling in beef, higher butterfat in milk, stronger eggshells.



3. Reducing Production Costs

- ❑ 60–70% of total livestock production is feed cost.
- ❑ Optimizing feed efficiency reduces waste.
- ❑ Use of feed additives (like enzymes and probiotics) enhances nutrient digestibility and cost-effectiveness.

4. Supporting Food Security

- ❑ Vital source of HBV protein, vitamins like B12, etc.
- ❑ Prevent malnutrition
- ❑ Proper animal nutrition ensures consistent food supply, even under climate stress or limited resources.

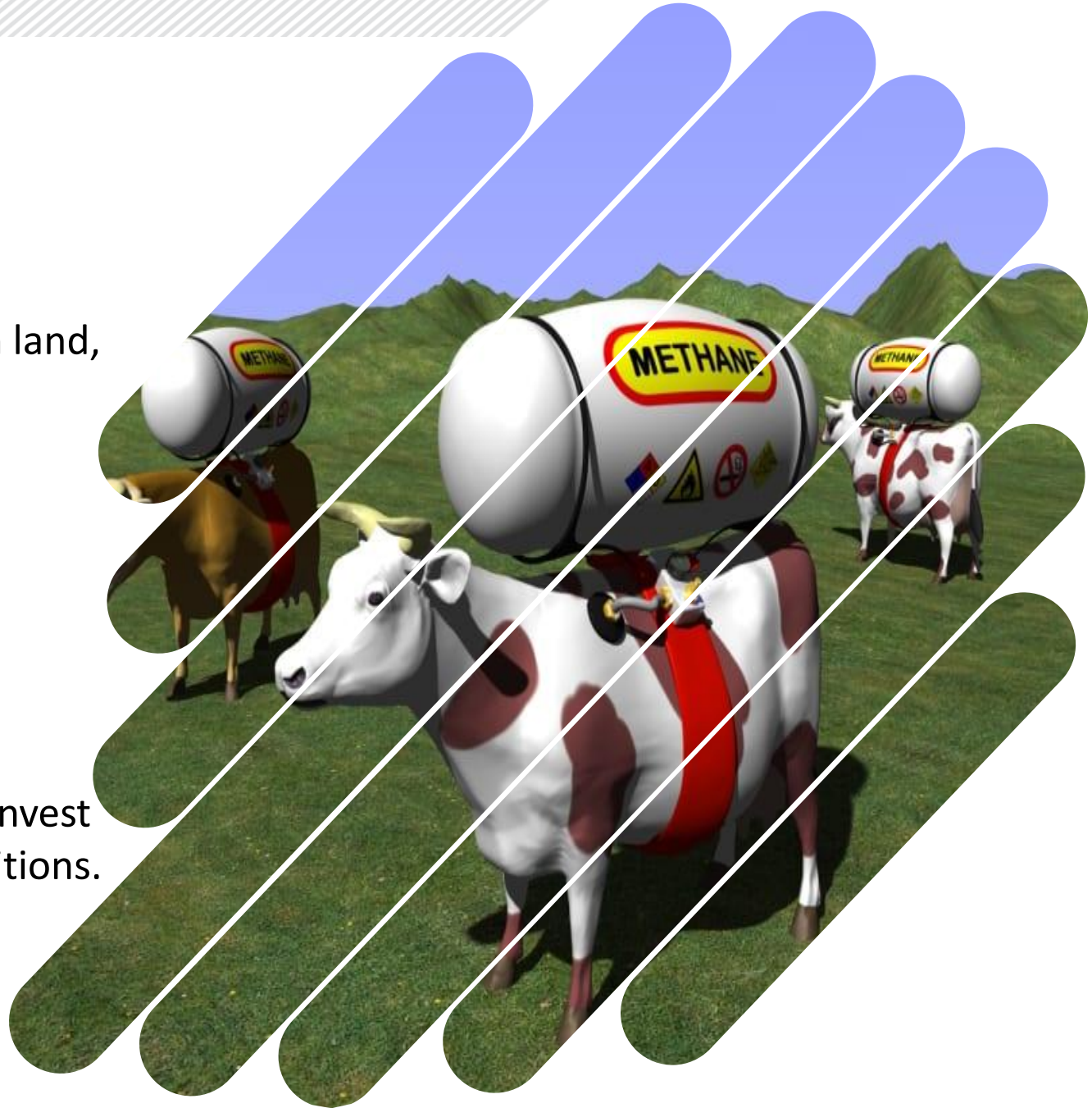


5. Minimizing Environmental Impact

- ❑ Reduce of methane emissions.
- ❑ Precision feeding contributes to climate-smart agriculture.
- ❑ Efficient animal production reduces pressure on land, water, and energy resources.

6. Promoting Sustainable Livelihoods

- ❑ Livestock farming supports rural livelihoods & income
- ❑ Good animal nutrition improves livestock profitability, helping farmers to earn more and invest in education, healthcare, and better living conditions.



7. Advancing Scientific and Technological Innovation

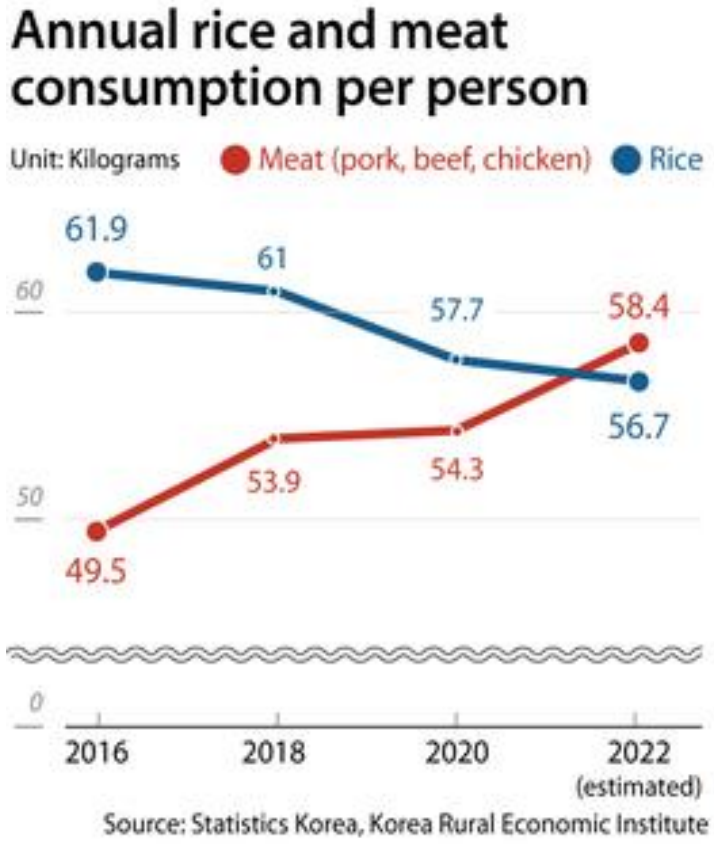
- ❑ Precision farming tools, and biotechnological advances (like microbiome engineering) are reshaping agriculture.
- ❑ Integration of data science like machine learning models for ration balancing into practical farming.

8. Safeguarding Public Health (One Health Approach)

- ❑ Reduce the need for antibiotics, lowering the risk of resistance (AMR).
- ❑ Control zoonotic diseases by supporting healthier, more resilient animals.
- ❑ Safe, nutritious livestock products contribute directly to human health outcomes.



- Consumption of foods of animal origin increases as economic status improves.
- In low-income nations, income elasticity of demand of animal protein is dependent on family income.
- Therefore, adequate supply of animal proteins will depend on the success of livestock research in animal nutrition and related disciplines.





1. Increasing Global Demand

•Rising global population (9 billion by 2050) is driving demand for animal protein.

•Economic growth in developing countries leads to dietary shifts from plant-based to animal-based foods.

2. Shift Toward High-Quality Protein

•Consumers are increasingly seeking lean meat, eggs, and dairy with better nutritional profiles.

•Demand is growing for animal products perceived to support health (e.g., omega-3 enriched eggs, probiotic yogurts).

3. Urbanization and Lifestyle Changes

•Urban dwellers typically consume more processed and ready-to-eat animal products (e.g., meat snacks, cheese, sausages).

•There's greater reliance on supermarkets and fast-food chains, influencing consumption patterns.

4. Regional Variations

•In Asia and Africa: rapid growth in poultry and dairy consumption.

•In developed countries: consumption of red meat is plateauing or declining, as poultry and plant-based rise.

5. Rise of Organic and Free-Range Products

Consumer awareness about animal welfare and sustainability is driving demand for:

- Organic meat, milk, and eggs.
- Free-range and grass-fed animal products.

6. Alternative Protein Sources

•Innovations in lab-grown meat, insect protein, and plant-based meat substitutes are emerging.

•These alternatives aim to address sustainability, ethics, and environmental concerns.

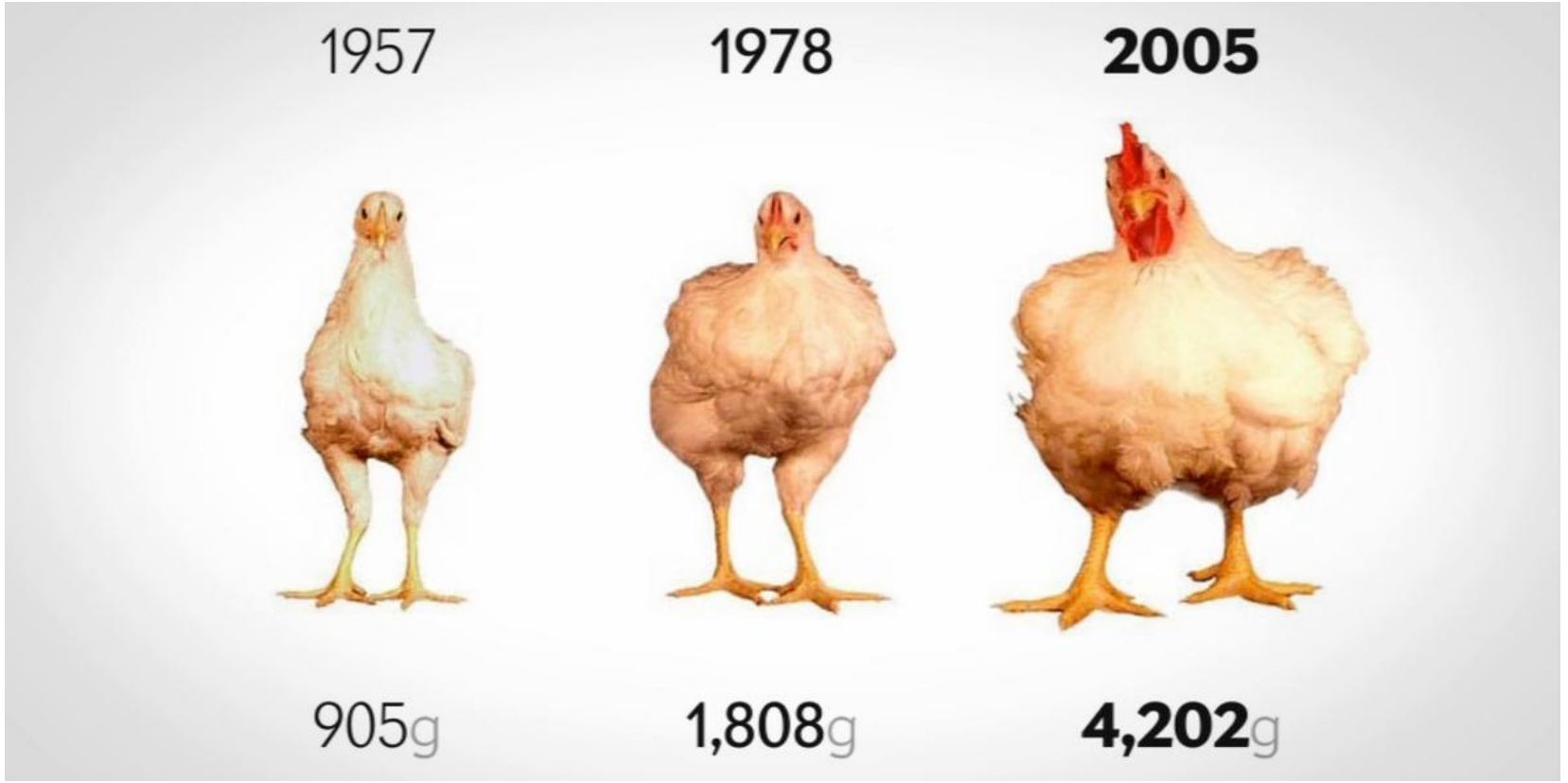
7. Impact of Global Crises

•Events like COVID-19 and climate-related disruptions have temporarily altered consumption patterns (e.g., reduced meat access, shift to local food sources).

Improved Feed Efficiency

- ❑ Selective breeding & improved nutrition have increase animal FCR.
- ❑ Implying less feed needed to achieve required production.

- ❑ However, despite the large advances within animal nutrition, there are many unknowns in this field



Dry Matter

- All the components of a feed excluding its water content. DM is in two categories, organic & inorganic and includes proteins, fats, carbohydrates (fiber, sugars, starch), Vitamins, minerals, etc.

DM (%) = 100 – Moisture (%)

- The major component of DM in most pasture grass & seeds is carbohydrate with exception of oilseeds.
- However, animals DM is mainly protein and lipids. Cell walls of plant are mainly carbohydrate (cellulose) and animals entirely proteins & lipids. Additionally, plants store energy in form of carbohydrates like starch while animals as lipids
- All animal requirements are expressed on dry matter basis as a standard unit.

Importance in nutrition

- Establishes the amount to nutrients available to the animal for health and production functions.
- Useful in ration formulation to avoid over/under feeding of nutrients to promote efficient nutrient use.

Dry Matter Intake [DMI]

DMI is the total amount of feed consumed by an animal on a dry matter basis.

DMI (Kg/day) = Total feed intake (kg) x DM%

Example: A lactating cow consumes 20 kg as-fed feed per day containing 60% moisture: Calculate the DMI.

DM = 100-60 = 40%: DMI = 20*0.4 = 8kg DM/d

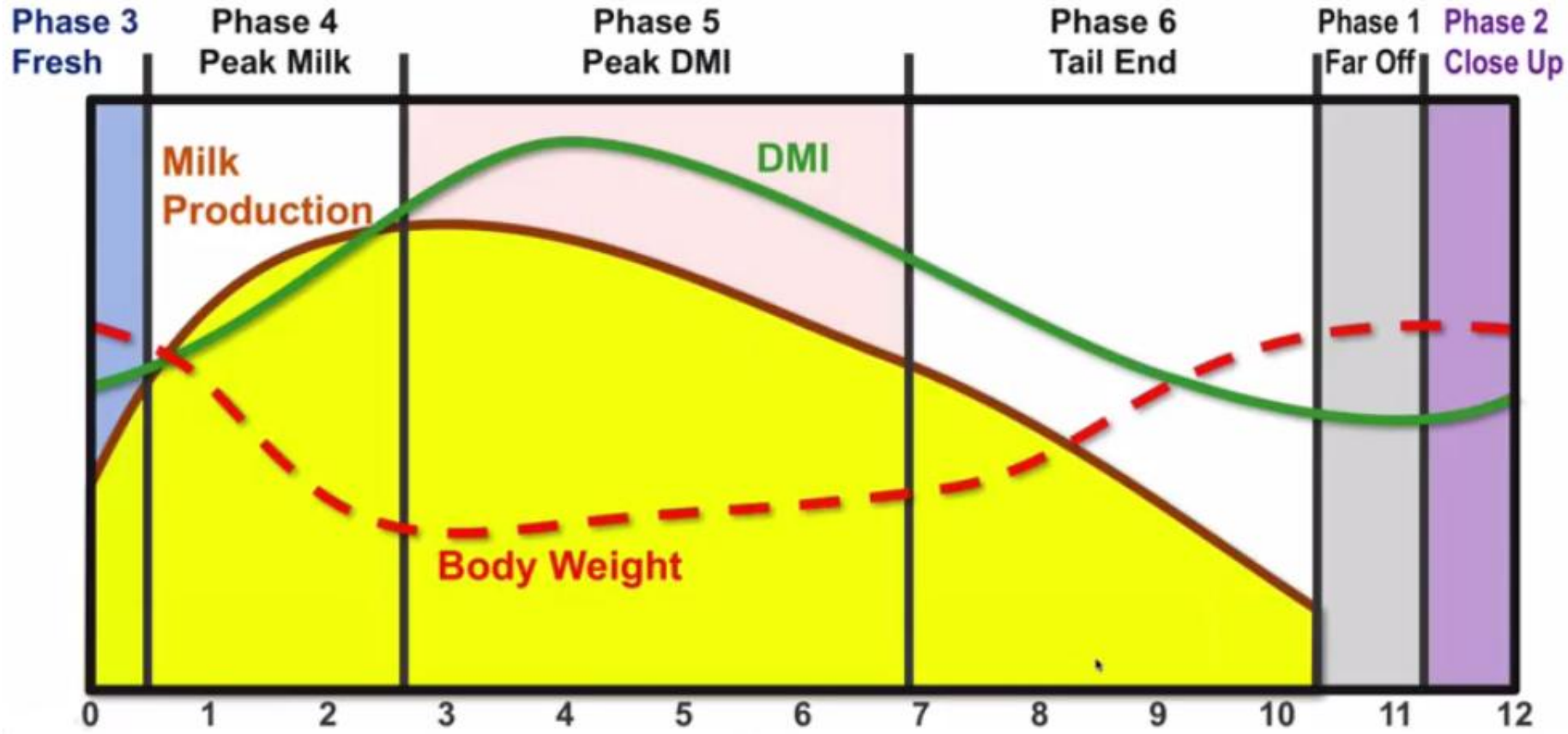
Lactating Cattle

Milk production = Energy expenditure : **While** Milk production peak = 4 – 8 weeks postpartum

DMI peak = 10 – 14 weeks postpartum

So,

Is milk production driven by intake **OR** intake is driven by milk production



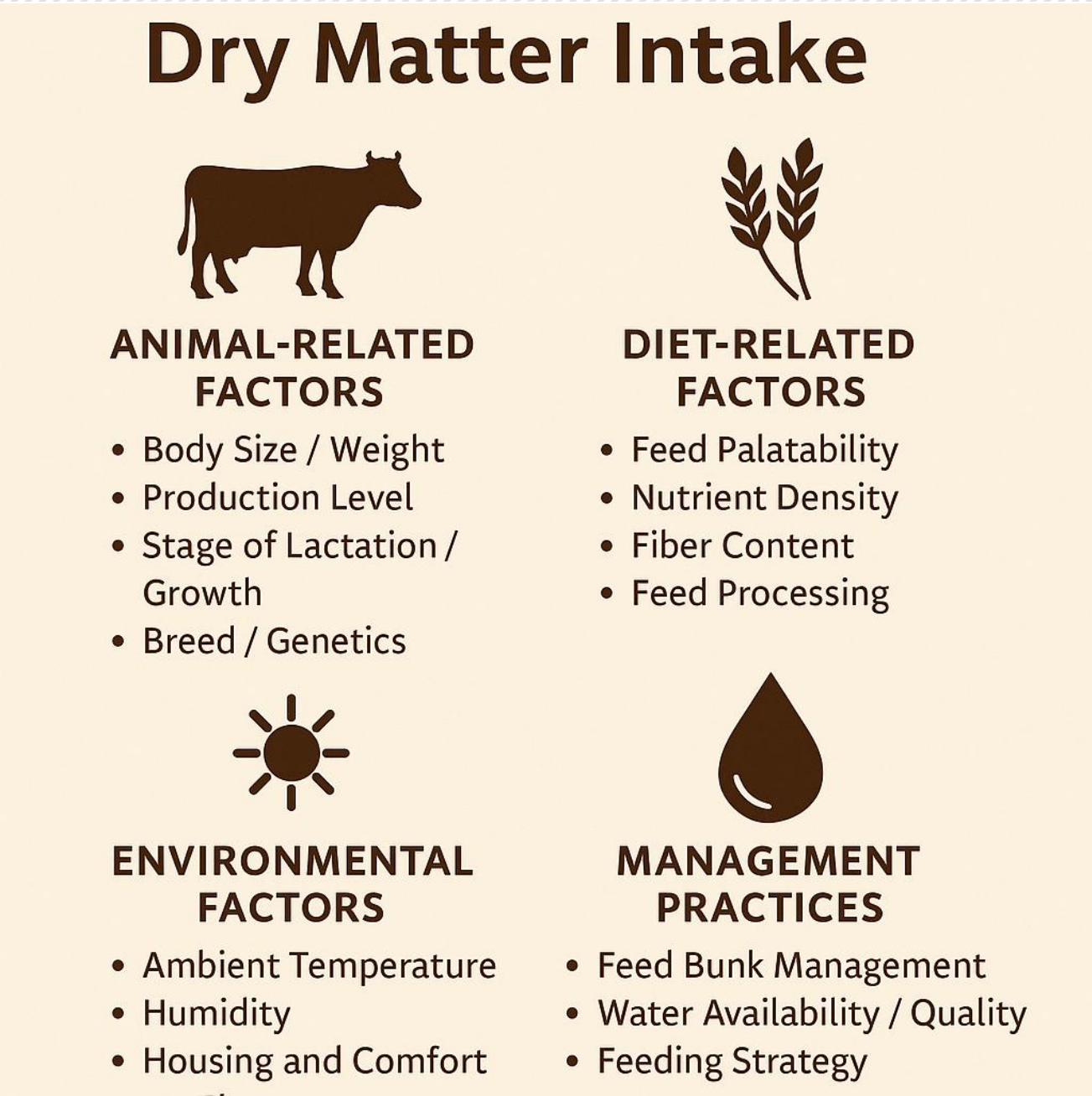
- Cows appear to consume feeds to meet their energy requirements. Therefore, energy intake is driven by milk production or (energy expenditure).

DMI is a function of milk production in response to energy expenditure: suggesting that DMI follows milk production



Primiparous Vs Multiparous Cows

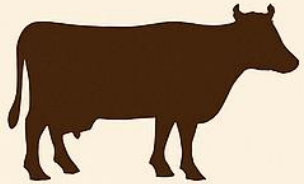
- Though DMI follows milk production in both, DMI for primiparous cows does not peak as early in lactation as do the multiparous.
- This is attributed to additional energy requirement for growth in primiparous cows
- This calls for separate feed regimen management factoring in the DMI and social hierarchy. Since primiparous cows are more timid and lower in hierarchy.
- Studies have shown more milk yield when managed separately.



Animal-Related Factors

- ✓**Body Size / Weight**: Larger animals have higher DMI due to greater maintenance requirements.
- ✓**Production Level**: High-producing animals (e.g., lactating cows) consume more feed to meet energy demands.
- ✓**Stage of Lactation / Growth**: Early lactation and rapid growth stages increase DMI.
- ✓**Breed / Genetics**: Some breeds inherently consume more due to metabolic differences.
- ✓**Health Status**: Illness, stress, or parasitism reduces appetite and feed intake.

Dry Matter Intake



ANIMAL-RELATED FACTORS

- Body Size / Weight
- Production Level
- Stage of Lactation / Growth
- Breed / Genetics



DIET-RELATED FACTORS

- Feed Palatability
- Nutrient Density
- Fiber Content
- Feed Processing



ENVIRONMENTAL FACTORS

- Ambient Temperature
- Humidity
- Housing and Comfort

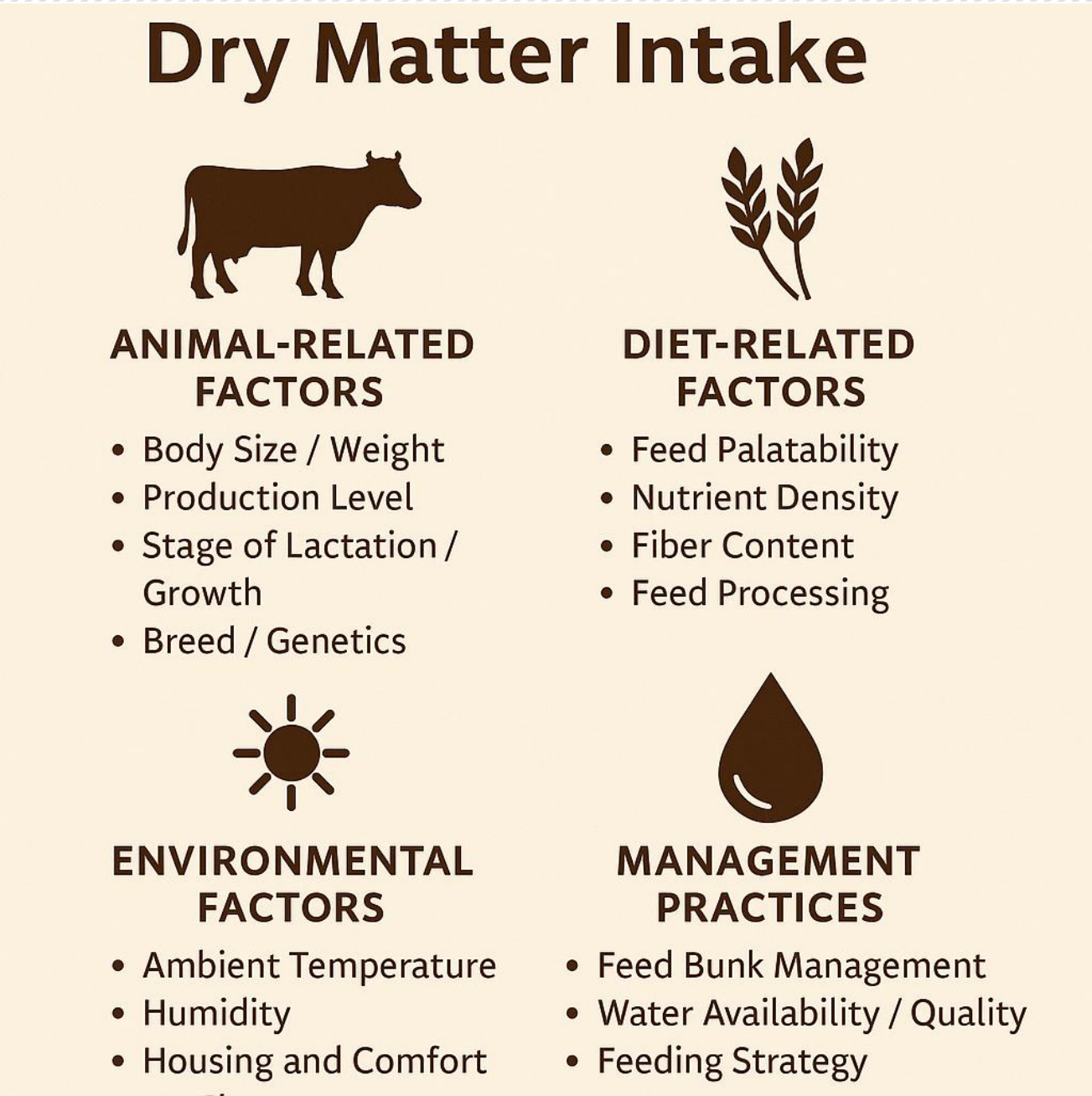


MANAGEMENT PRACTICES

- Feed Bunk Management
- Water Availability / Quality
- Feeding Strategy

Diet-Related Factors

- ✓**Feed Palatability**: Taste, texture, and smell influence voluntary intake.
- ✓**Nutrient Density**: High-energy, nutrient-dense feeds may reduce overall DMI as requirements are met faster.
- ✓**Fiber Content**: High fibre (especially indigestible fiber) reduces DMI by increasing gut fill.
- ✓**Feed Processing**: Grinding, pelleting, or ensiling can improve digestibility and intake.



❖ Feed moisture content: High MC tends to limit DMI

most high MC feeds are fermented and the decrease in DMI is due to fermentation end products not water.

❖ Feed fiber content: high fiber tends to limit DMI

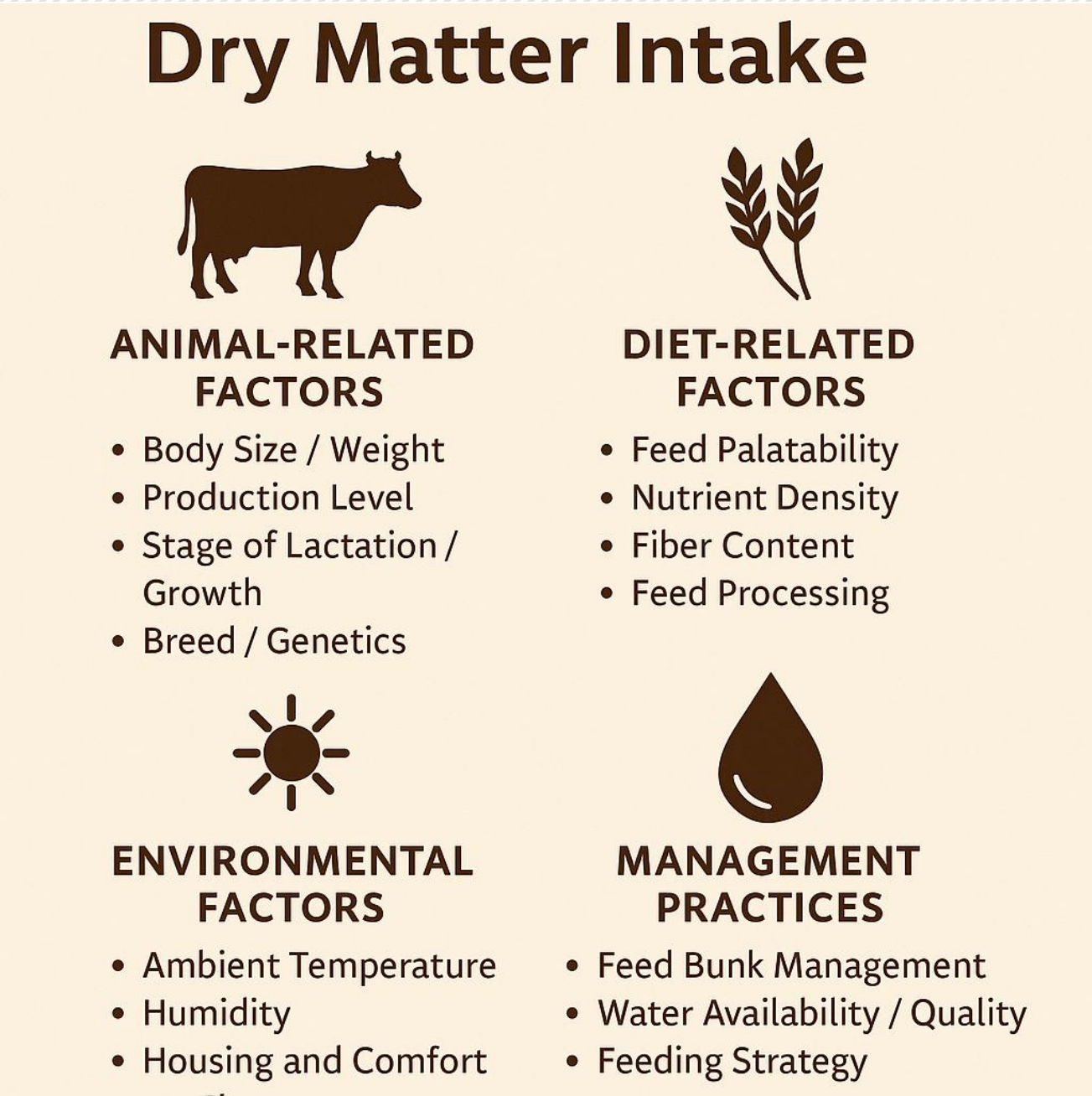
At high fiber concentration, rumen fill tends to limit DMI

At low neutral detergent fiber concentration, energy feedback mechanism tends to limit DMI.

❖ Ratio of forage to concentrate (F:C)

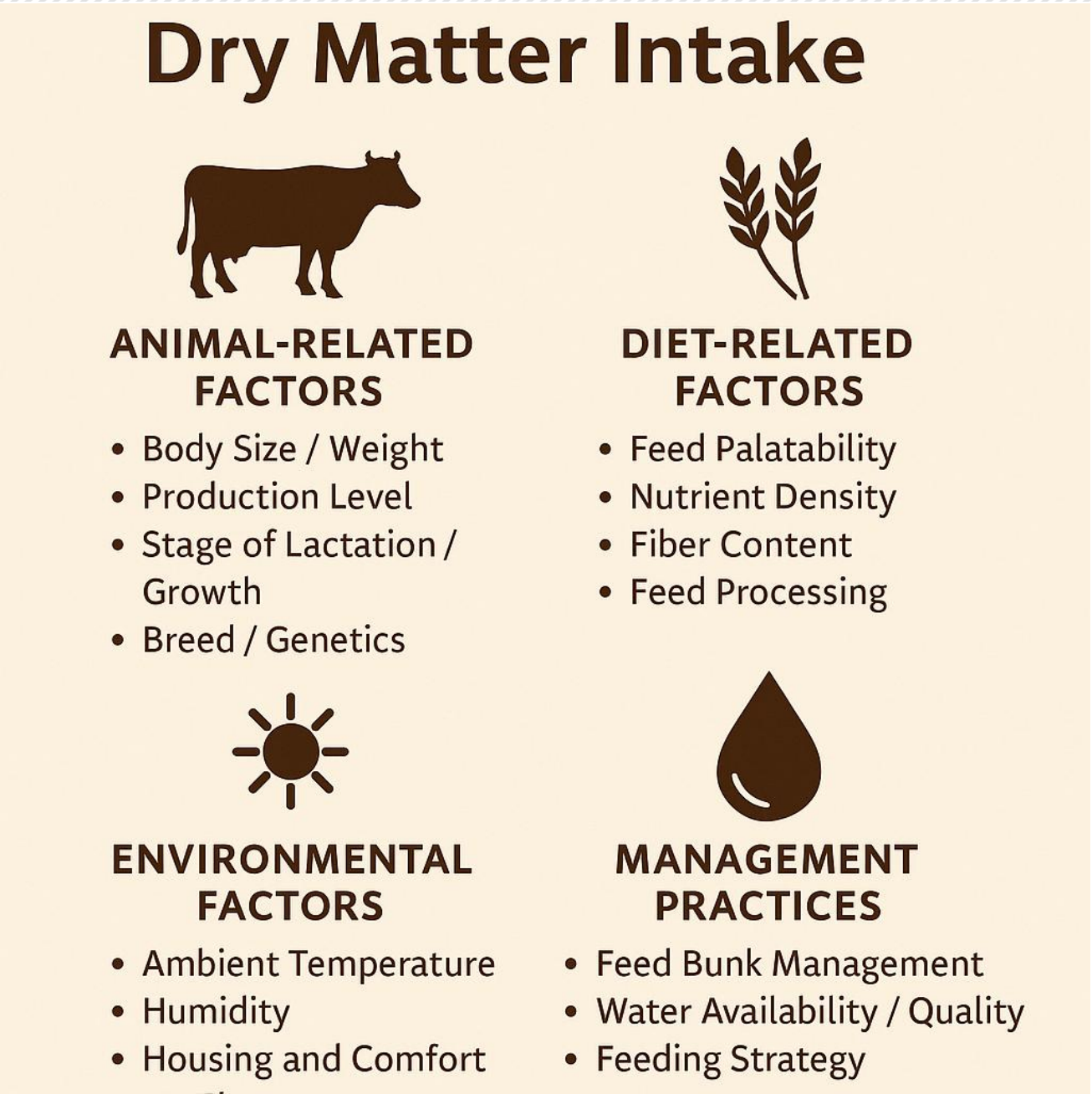
High concentrate supplementation increases DMI due to

✓ Increased rate of passage
✓ Higher digestibility
✓ Physical characteristics associated



Environmental Factors

- ✓**Ambient Temperature**: Heat stress reduces DMI, while cold stress may increase it.
- ✓**Humidity**: High humidity can worsen heat stress and reduce intake.
- ✓**Housing and Comfort**: Poor ventilation, overcrowding, and lack of water reduce feeding behavior.
- ✓**Feeding Frequency**: Frequent feeding or feed availability throughout the day encourages higher intake.

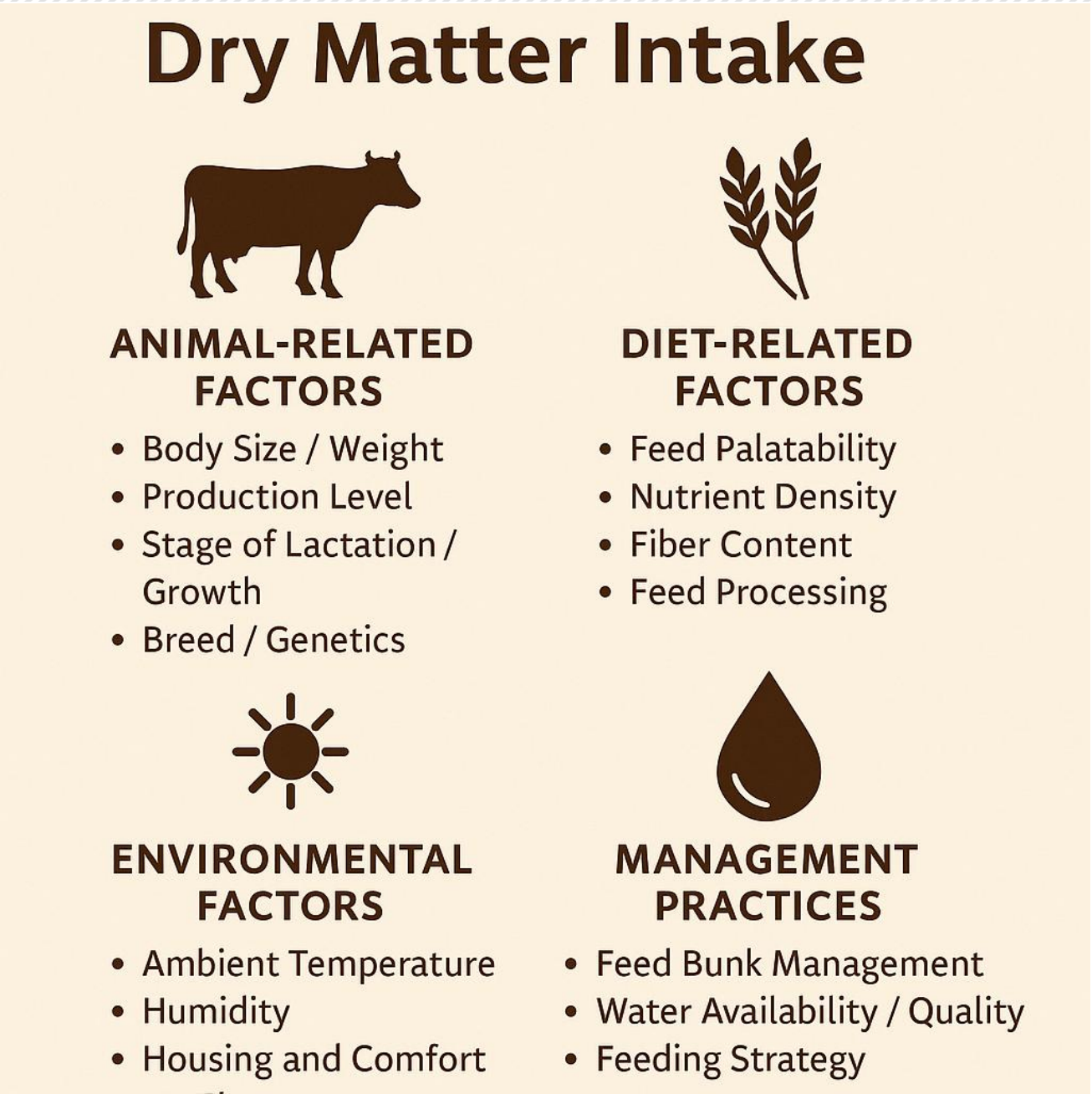


❖ Temperature:

The thermal neutral zone of dairy cattle is about 5^0-20^0C (lower critical – upper critical temperature). Temperatures out of these zone alter intake & metabolic activities.

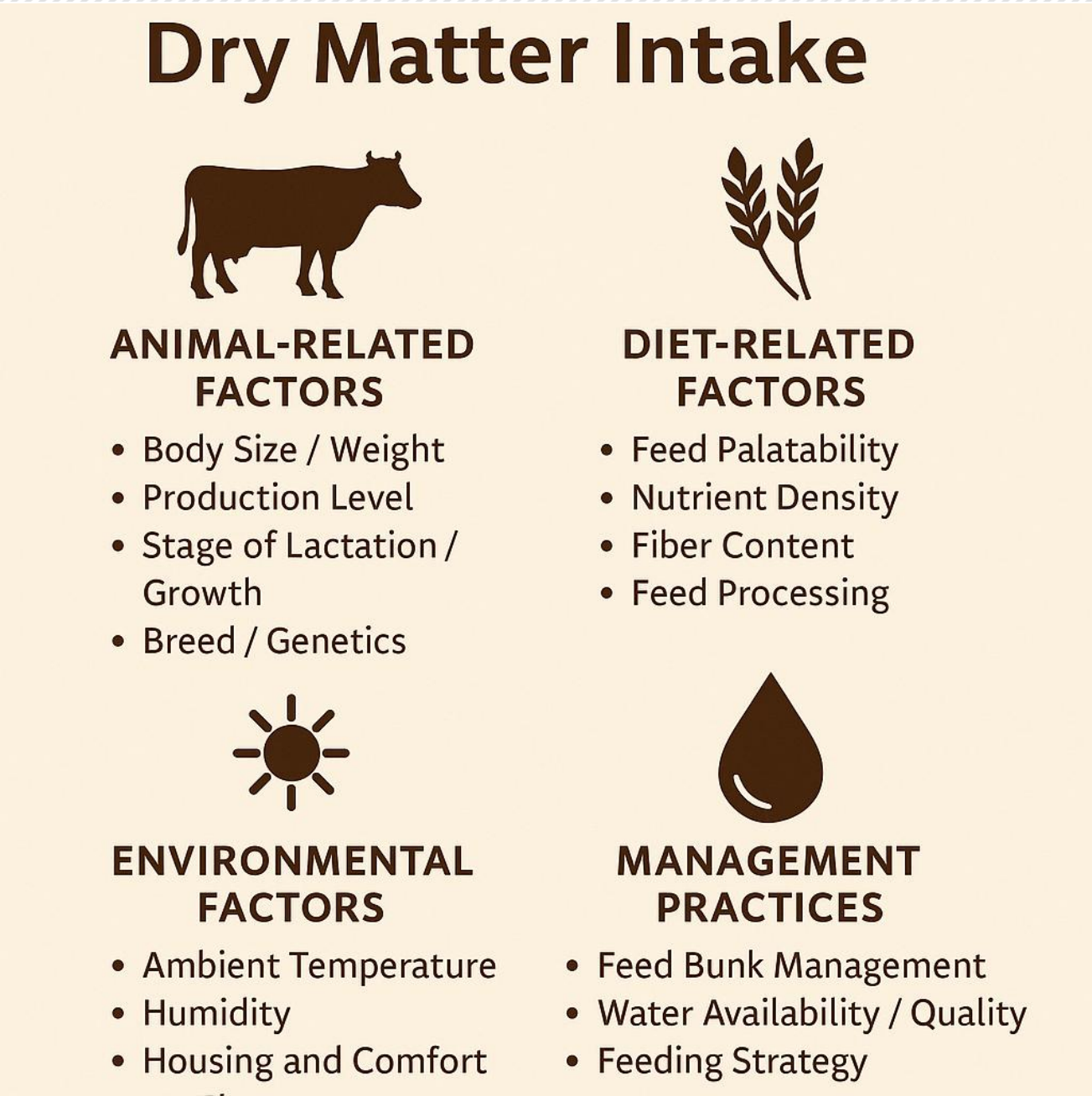
- ✓ Cold conditions increase insulation, DMI, and basal metabolism, rumination, reticulo-rumen motility, and rate of passage.
- ✓ In extreme cold, DMI does not increase with metabolism. Since animals are in negative energy balance, they shift energy use from productive purpose to heat production.
- ✓ Higher temperatures decrease milk production due to decreased DMI.

This implies that smaller bodied animals like Jersey may be more productive that large frame Friesians in the tropics.



Management Practices

- ✓**Feed Bunk Management**: Clean, accessible, and well-maintained bunks promote intake.
- ✓**Water Availability and Quality**: Water is essential for digestion; poor quality or limited access lowers DMI.
- ✓**Feeding Strategy**: TMR (Total Mixed Ration) vs. individual ingredients affects uniformity and intake.



Feeding Methods

- ✓ TMR increase intake
- ✓ Increased feeding frequency allows increased milk production: just increasing intake from 2 to 4 times a day increases Av. Daily gain in beef cattle by 16%
- ✓ Sequence of feeding affects DMI: Giving highly fermentable carbohydrates could cause acidosis lowering pH and DMI: Giving forages first in the morning could lead to formation of a fiber mat in the rumen and provide buffering capacity from the forage & increase salivation.
- ✓ Competition for feed: accessibility to feed must be more than 5hrs of actual eating time per day.

Equations for Predicting DMI

In the simplest case, DMI is related to wgt [DMI = 3% BW]

But it ranges from 2-4% of BW

Physiological State

The dictates level of DMI

Dairy animals in early lactation consume less: This lag is attributed to hormonal imbalance especially

- ✓ thyroxine,
- ✓ Thyrotropin-releasing hormone (TRH)
- ✓ Prolactin/ luteotropic hormone: LTH, leptin and oxytocin because of gestation.

Dairy cows

Advanced lactation = consume 32g/Kg BW

Early lactation = consume 28g/Kg BW

Growing heifers = consume 22g/Kg BW

Dairy Animals

In lactating animals, DMI is related to milk yield (Y)

Net energy (NE_L) of lactation

$$\text{DMI (Kg/d)} = \frac{NE_L \text{ required (MJ)}}{NE_L \text{ concentration of diet (MJ/Kg)}}$$

Grower/ Beef Cattle

DMI is related to body weight gain

$$\text{DMI (Kg/d)} = \frac{NE_m + NE_g}{NE_{diet}}$$

Equations for Predicting DMI

National Research Council (NRC, 2001)

The NRC prediction model is based on the principle that animals consume to satisfy:

- ✓ Energy requirements at a given level of butter fat production
- ✓ Metabolic body weight
- ✓ Days in milk (DIM) expressed as Week of Lactation (WOL)

Why FCM

- ✓ Fat % is a good measure of energy expenditure
- ✓ Though average is 4%, butter fat varies with animal
- ✓ Therefore, standard FCM (4% FCM) permits comparison of animals on a common energy basis.
- ✓ It adjusts the actual milk yield to the 4% butter fat standard

Equation/ Model

$$DMI\ (Kg/d) = 0.372*FCM + 0.0968*BW^{0.75} * (1-e^{(-0.192*(WOL+3.67))})$$

Where:

- ✓ FCM = Fat corrected milk (Kg/d) standardized to 4%
- ✓ 4% FCM = (0.4*Y(kg/d)) + 15*Fat (Kg/d) **OR** 0.4(Y)+15(Y*FAT%)

- ✓ $BW^{0.75}$ = Metabolic body weght (Kg)
- ✓ WOL = Week of lactation

- ✓ $(1-e^{(-0.192*(WOL+3.67))})$ = adjusts for despressed DMI during early lactation
- ✓ Constant e = Euler's number [exceptional function which is the base of natural algorithm] = Approx. 2.718

DMI PRACTICAL PREDICTION

$$DMI\ (Kg/d) = 0.372*FCM + 0.0968*BW^{0.75} * (1-e^{(-0.192*(WOL+3.67))})$$

Production Factor (0.372 × FCM):

- ❑ Reflects the feed requirement for milk production. Higher milk yield → higher DMI.

Maintenance Factor ($0.0968 \times BW^{0.75}$):

- ❑ Accounts for the cow's metabolic weight and the basic nutritional needs to maintain body functions.

Lactation Curve Modifier ($1-e^{(-0.192*(WOL+3.67))}$)

- ❑ Adjusts intake based on stage of lactation.
- ❑ Early lactation (low WOL): intake is limited.
- ❑ As WOL increases, DMI gradually rises and then plateaus

This equation is particularly helpful for **formulating rations in dairy cows**, especially:

- ✓ During early lactation when energy demand is high.
- ✓ For precision feeding and avoiding under/overfeeding.

Class Exercise I

Using the NRC (2001) DMI prediction model, compute the DMI of a 500 Kg Friesian cow in her 40th week of lactation producing 30 kg of milk daily with a butter fat content of 4.6%. **[5 minutes]**

1. Definition

- Metabolic body weight (MBW) is body weight raised to the power of 0.75:
- $MBW=BW^{0.75}$ where BW = body weight (kg).
- It is used to describe the metabolically active size of an animal, rather than just its live weight.

Why 0.75?

- If you plotted energy use (heat production, oxygen consumption, feed intake, etc.) vs. body weight across many species (mice, cows, elephants), the relationship is not linear.
- In the 1930s, Max Kleiber studied this and found that metabolism scales with body weight to the ¾ power.
  - ✓ A mouse eats much more feed per kg of body weight than a cow.
  - ✓ But when you adjust by $BW^{0.75}$, energy needs across species become comparable.

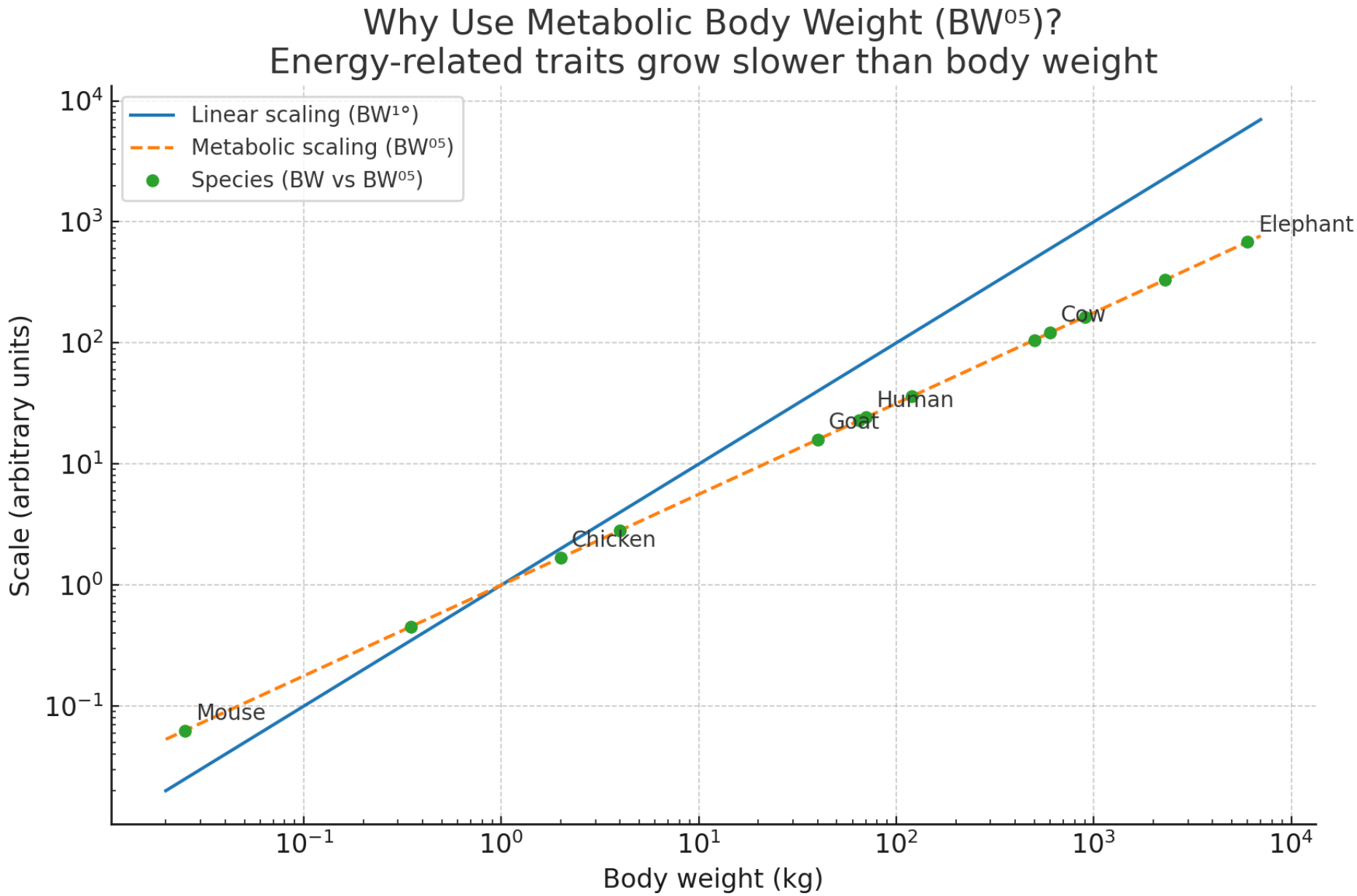
This scaling law is called Kleiber's Law.

Biological Meaning

- Larger animals have more body tissue, but not all tissue uses energy equally.
- Small animals have higher surface area-to-volume ratios, so they lose heat faster and need relatively more energy for maintenance.
- The exponent 0.75 captures how metabolic rate increases with size but at a diminishing rate.

Uses in Animal Science

- Maintenance energy requirement: expressed per unit of MBW (MJ or kcal per $kgBW^{0.75}$).
- Protein turnover, water needs, feed intake: often scale with MBW.
- Comparisons across species: MBW allows nutritionists to compare poultry, goats, cattle, and even lab rats on the same metabolic scale.


Why Use Metabolic Body Weight (BW⁰⁵)?
Energy-related traits grow slower than body weight
Linear scaling (BW¹°)
Metabolic scaling (BW⁰⁵)
Species (BW vs BW⁰⁵)
Mouse
Chicken
Goat
Human
Cow
Elephant
Scale (arbitrary units)
Body weight (kg)
10⁴
10³
10²
10¹
10⁰
10⁻¹
10⁻¹
10⁰
10¹
10²
10³
10⁴

$$DMI\ (Kg/d) = 0.372*FCM + 0.0968*BW^{0.75} * (1-e^{(-0.192*(WOL+3.67))})$$

Fat Corrected Milk

FCM = (0.4*Y(kg/d)) + 15*fat(kg/d)
= (0.4*30) + 15(4.6%*30)
= 32.7 kg/d

Metabolic Body Weight

$BW^{0.75} = 500^{0.75}$
= 105.74 Kg

Lactation Curve Factor $(1-e^{(-0.192*(WOL+3.67))})$

$= (1-e^{(-0.192*(40+3.67))}) = (1-e^{(-0.192*43.67)})$
$= (1-2.718^{(-8.38)}) = (1-0.000229)$
= 0.9998

Therefore;

DMI = 0.372*32.7 + 0.0968*105.74 * 0.9998
= 22.4

DMI = 22.4 Kg per day

Thumb Rules

DMI Requirements

- ❑ Cow: 2.5% Body weight
- ❑ Buffalo: 3.0% Body weight
- ❑ Beef: 2.0-2.7% Body weight
- ❑ Sheep: 3.5-4.5 Body weight

Roughage DMI

During seperate feeding

- ❑ 2/3 of DMI
- ✓ Wet/ green roughage intake - 2/3
- ✓ Dry roughage intake - 1/3

Concentrate DMI

- ❑ 1/3 of DMI

Using TMR

All DMI should come from the ration.

Class Exercise I

Given cow has a body weight of 500kg:

a) Calculate its total DMI
b) Calculate its roughage intake (Green and dry)
c) Calculate its concentrate intake